



JAVA PROGRAMMING

(340)

NATIONAL – 2019

Production Portion:

Program 1: Traffic Light _____ (345 points)

TOTAL POINTS _____ (***345 points***)

**Graders: Please double check and verify all scores
and answer keys!**

Property of Business Professionals of America.
May be reproduced only for use in the Business Professionals of America
Workplace Skills Assessment Program competition.



You will have ninety (90) minutes to complete your work.

Your name and/or school name should *not* appear on work you submit for grading.

1. Create a folder on the flash drive provided using your contestant number as the name of the folder.
2. Copy your entire solution/project into this folder.
3. Submit your entire solution/project so that the graders may open your project to review the source code.
4. Ensure that the files required to run your program are present and will execute on the flash drive provided.

*Note that the flash drive letter may *not* be the same when the program is graded as it was when you created the program.

*It is recommended that you use relative paths rather than absolute paths to ensure that the program will run regardless of the flash drive letter.

The graders will *not* compile or alter your source code to correct for this.
Submissions that do *not* contain source code will *not* be graded.

Assumptions to make when taking this assessment:

- The traffic light will have 3 round lights (red, yellow, green) and a rectangular casing

Development Standards:

- Your Code must use a consistent variable naming convention.
- All subroutines (if any), functions (if any), and methods (if any) must be documented with comments explaining the purpose of the method, the input parameters (if any), and the output (if any).
- If you create a class, then you must use Javadoc comments.



Your application will be graded on the following criteria:

Solution and Project

The project is present on the flash drive	10 points
The projects main class is named TrafficLight	10 points
The projects helper class is named TrafficLightPanel	10 points
The projects helper class is named TrafficControlPanel	10 points

Program Execution

The program runs from the USB flash drive	_____	15 points
---	-------	-----------

If the program does *not* execute, then the remaining items in this section receive a score of zero.

The program displays the stop light with the button activated	_____	30 points
The program displays the red light in the correct position and color	_____	20 points
The program displays the yellow light in the correct position and	_____	20 points
The program displays the green light in the correct position and color	_____	20 points
The program displays the stoplight casing in an appropriate color around the	_____	20 points
The program changes the light position and color when the button is	_____	30 points
The program displays the stop light at an appropriate size for JPanel	_____	10 points

Source Code Review

The source code is properly commented		
A comment containing the contestant number is present	_____	10 points
Methods and code sections are commented	_____	20 points
A method exists to set the panel layout	_____	20 points
A method exists to set the dimensions of the traffic light	_____	20 points
A method exists to paint the traffic light	_____	20 points
Code extends JPanel for TrafficLightPanel class	_____	20 points
Code extends JPanel for TrafficControlPanel class	_____	20 points



Code uses a consistent variable naming convention

10 points

345

Total Points =



Suggested Solution

```
/**
 * BPA Java Programming Contest : Team Directory
 *
 * @author - contestant number goes here
 *
 * @version January 2018
 */

import javax.swing.*;

public class TrafficLight
{
    //-----
    // Creates and presents the program frame.
    //-----
    public static void main (String[] args)
    {
        JFrame lightFrame = new JFrame ("Traffic Light");
        lightFrame.setDefaultCloseOperation (JFrame.EXIT_ON_CLOSE);

        lightFrame.getContentPane().add (new TrafficControlPanel());

        lightFrame.setSize(160,250);
        lightFrame.show();
    }
}
```

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class TrafficControlPanel extends JPanel
{
    TrafficLightPanel light;

    public TrafficControlPanel()
    {
        setLayout(new BorderLayout());
        light = new TrafficLightPanel();
        add(light, BorderLayout.CENTER);

        JButton change = new JButton("Change Light");
        change.addActionListener(new ChangeListener());
        add(change, BorderLayout.SOUTH);
    }
}
```



```
class ChangeListener implements ActionListener
{
    public void actionPerformed(ActionEvent event)
    {
        light.change();
    }
}
```

```
import javax.swing.*;
import java.awt.*;
```

```
public class TrafficLightPanel extends JPanel
{
    private int currentState = 0;
    private final int NUM_LIGHTS = 3;
    private final int X = 50, Y = 10, WIDTH = 50, HEIGHT = 130;
    private final int DIAMETER = 30;
    private final int X_OFFSET = 10, Y_OFFSET = 10;
    private final int PANEL_WIDTH = 150, PANEL_HEIGHT = 230;
    //-----
    // Creates the traffic light panel
    //-----
    public void TrafficLightPanel()
    {
        setPreferredSize(new Dimension(PANEL_WIDTH, PANEL_HEIGHT));
    }

    public Dimension getPreferredSize()
    {
        return new Dimension(PANEL_WIDTH, PANEL_HEIGHT);
    }

    public void paintComponent(Graphics page)
    {
        super.paintComponent(page);

        int lightOn = currentState % NUM_LIGHTS;

        setBackground(Color.white);

        page.setColor(Color.lightGray);
        page.fillRect(X, Y, WIDTH, HEIGHT);

        if (lightOn == 0)
            page.setColor(Color.red);
        else
            page.setColor(Color.darkGray);
    }
}
```



```
    page.fillOval(X+X_OFFSET, Y+Y_OFFSET, DIAMETER, DIAMETER);

    if (lightOn == 1)
        page.setColor(Color.yellow);
    else
        page.setColor(Color.darkGray);
    page.fillOval(X+X_OFFSET, Y+DIAMETER+2*Y_OFFSET, DIAMETER, DIAMETER);

    if (lightOn == 2)
        page.setColor(Color.green);
    else
        page.setColor(Color.darkGray);
    page.fillOval(X+X_OFFSET, Y+2*DIAMETER+3*Y_OFFSET, DIAMETER, DIAMETER);

}

public void change()
{
    currentState++;
    repaint();
}
}
```